

(1) $\pi_{\text{Grade}}(\sigma_{\text{Ssex}='女' \wedge \text{Cname}='OS'}(S \bowtie SC \bowtie C))$ (2 分)

RANGE C CX (2 分)

RANGE S SX

GET W(SC.Grade): $\exists SX \exists CX (SX.Sno=SC.Sno \wedge SC.Cno=CX.Cno \wedge CX.Cname='OS' \wedge SX.Ssex='女')$

(2)

SC SCX

SC SCY

$\pi_{\text{Sno,Sname}}(S \bowtie \pi_{\text{Sno}}(\sigma_{\text{SCX.Sno}=\text{SCY.Sno} \wedge \text{SCX.Cno} \neq \text{SCY.Cno}}(\text{SCX} \times \text{SCY})))$ (2 分)

RANGE SC SCX

RANGE SC SCY

GET W(S.Sno,S.Sname): $\exists SCX \exists SCY (\text{SCX.Sno}=S.Sno \wedge \text{SCY.Sno}=S.Sno \wedge \text{SCX.Cno} \neq \text{SCY.Cno})$ (2 分)

(3)

$\pi_{\text{Sno,Cno}}(SC) \div \pi_{\text{Cno}}(C) \bowtie \pi_{\text{Sno,Sname}}(S)$ (2 分)

RANGE C CX (2 分)

RANGE SC SCX

GET W(S.Sno,S.Sname): $\forall CX (\exists SCX (\text{SCX.Cno}=CX.Cno \wedge \text{SCX.Sno}=S.Sno))$

三、SQL 语言 (共 18 分)

(1) (3 分)

Alter table dept add constraint dnameC dname unique;

Alter table emp add constraint fkmgr foreign key mgr references emp(empno);

Alter table emp add constraint salC check(sal between 0 and 100000);

(2) (3 分)

Select * from emp

where sal < 3000 and deptno in (select deptno from dept where dname='IT');

(3) Select * from emp where sal > all (select sal from emp where deptno in (select deptno from dept where dname='IT' or dname='Business')); (3 分)

(4)

(3 分)

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Select ename, empno, sal from emp
      where ename like '刘%' and deptno in
      (select deptno from dept group by deptno having count(*)=(select max(count(*)) from
deptno group by deptno));
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(5) Create view dSal as select deptno, avg(sal) avgSal from emp group by deptno; (3 分)

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Select * from emp where sal > (select avgSal from dSal where emp.deptno=dSal.deptno);
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(6) Select deptno, count(*) eNum, avg (sal) avgSal, max(sal) maxSal,min(sal) minSal from emp
group by deptno order by avgSal desc;